

Transformer-Based Chinese News Classification

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Abstract

This report presents two experiments on Transformer-based Chinese news classification. The model was trained and evaluated under two settings. Task A used the original text directly, while Task B introduced text preprocessing before training and testing. The comparison focuses on the effect of preprocessing on validation performance, final test performance, and representative misclassified samples.

The results show that Task B achieved a small but consistent improvement over Task A. The final test accuracy increased from 83.96% to 84.20%, and the test loss decreased from 0.55 to 0.52. Misclassified-sample analysis further shows that preprocessing can help the model focus on core lexical cues in some headlines, but it can also remove useful punctuation or structural information in other cases. Overall, the experiment suggests that lightweight preprocessing is beneficial on average, although its effect remains limited.

1 Introduction

This project runs and evaluates a Transformer-based classifier for Chinese news text classification. The experiment includes two settings: one using the original text and one using preprocessed text. The report also presents the final test loss and accuracy, provide running evidence, output misclassified documents, and analyze representative cases where preprocessing helps or hurts the model.

This report therefore focuses on the experimental setup, quantitative comparison, and qualitative error analysis rather than on the internal structure of Transformer itself. The main question is whether a simple preprocessing strategy can improve classification performance under the same model architecture and training conditions.

2 Experimental Setup

2.1 Task Definition and Dataset

The dataset used in this project is the truncated version of **THUCNews**. The classification task is to assign each Chinese news sentence to one of ten categories: Finance, Real Estate, Stocks, Education, Technology, Society, Politics, Sports, Game, and Entertainment.

According to the running logs, the experiment used approximately 50,000 samples in total, split into 45,000 training samples, 2,500 validation samples, and 2,500 test samples.

2.2 Task A and Task B

The two required experimental settings are defined as follows:

- **Task A:** the original text is used directly without additional preprocessing.
- **Task B:** the input text is preprocessed before tokenization and model training.

In Task B, the preprocessing pipeline includes three operations:

- removing leading and trailing whitespace,
- normalizing repeated whitespace,
- removing common Chinese and English punctuation.

After preprocessing, the text is still tokenized at the character level. Therefore, the comparison between Task A and Task B isolates the effect of text preprocessing rather than changing the model design.

2.3 Implementation and Hyperparameters

The experiment was conducted with the provided Transformer implementation. The main hyperparameters are summarized in Table 1.

Table 1: Main hyperparameters used in the experiment.

Parameter	Value
Epochs	6
Batch size	128
Pad size	32
Learning rate	5×10^{-4}
Embedding dimension	300
Hidden size	1024
Attention heads	5
Encoder layers	2
Dropout	0.5

The two runs correspond to:

`TASK_NAME=A python3 run.py`

`TASK_NAME=B python3 run.py`

In both settings, the model was trained for six epochs, evaluated on the validation set after each epoch, and then tested on the test set. Misclassified test samples were also exported for later comparison.

The training code files were run on an NVIDIA RTX 4090 GPU (24 GB). The screenshots generated during the two runs are shown in Figures 1–4.

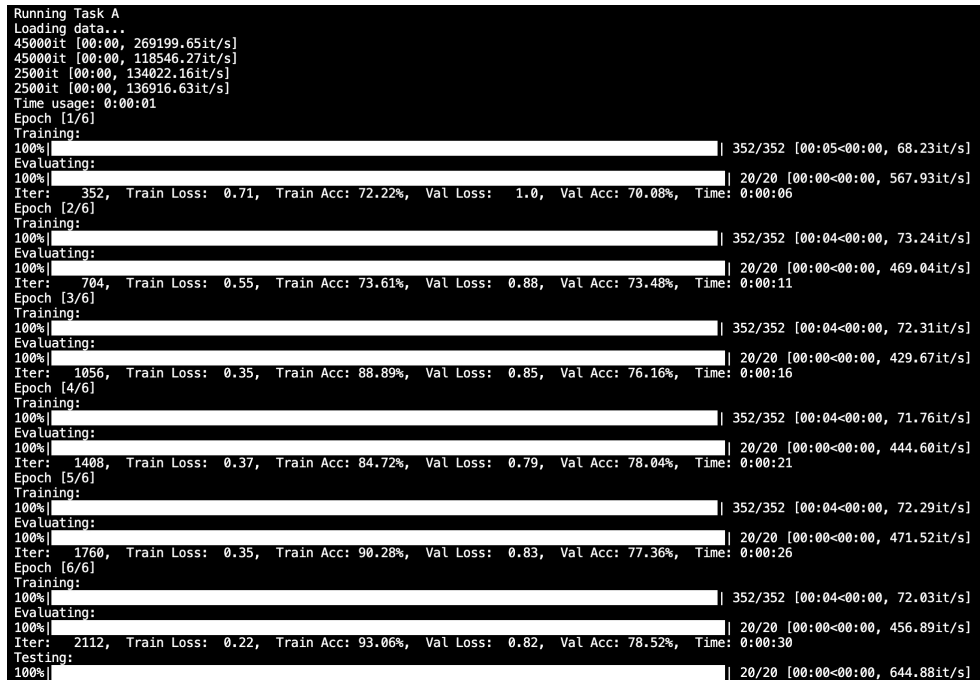


Figure 1: Running output screenshot for Task A.

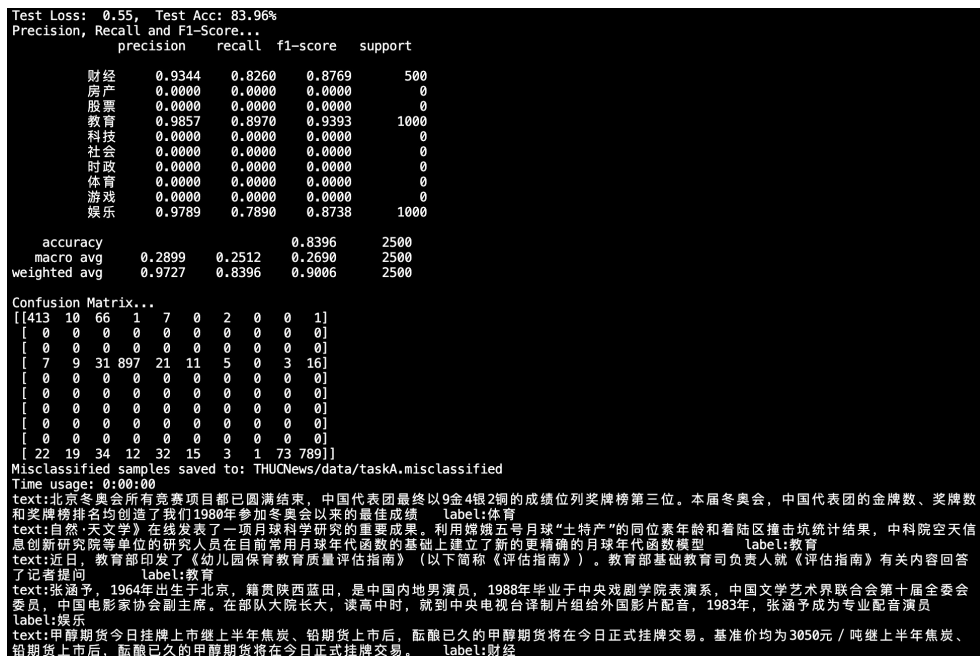


Figure 2: Validation / accuracy figure for Task A.

Figure 3: Running output screenshot for Task B.

```

Test Loss: 0.52, Test Acc: 84.20%
Precision, Recall and F1-Score...
precision    recall  f1-score   support

 财经      0.9624    0.7680    0.8543     500
  房产      0.0000    0.0000    0.0000       0
  股票      0.0000    0.0000    0.0000       0
  教育      0.9859    0.9090    0.9459    1000
  科技      0.0000    0.0000    0.0000       0
  社会      0.0000    0.0000    0.0000       0
  时政      0.0000    0.0000    0.0000       0
  体育      0.0000    0.0000    0.0000       0
  游戏      0.0000    0.0000    0.0000       0
  娱乐      0.9795    0.8120    0.8879    1000

 accuracy          0.8420     2500
  macro avg       0.2928    0.2489    0.2688     2500
weighted avg       0.9786    0.8420    0.9044     2500

Confusion Matrix...
[[384  12  87  1  8  2  0  0  2  4]
 [  0  0  0  0  0  0  0  0  0  0]
 [  0  0  0  0  0  0  0  0  0  0]
 [  6  7  21 900  10  15  6  0  13  13]
 [  0  0  0  0  0  0  0  0  0  0]
 [  0  0  0  0  0  0  0  0  0  0]
 [  0  0  0  0  0  0  0  0  0  0]
 [  0  0  0  0  0  0  0  0  0  0]
 [  9 28 43 12 23 21  4  5 43 812]]
Misclassified samples saved to: THUNews/data/taskB.misclassified
Time usage: 0:00:00
text:北京冬奥会所有竞赛项目都已圆满结束，中国代表团最终以9金4银2铜的成绩位列奖牌榜第三位。本届冬奥会，中国代表团的金牌数、奖牌数和奖牌榜排名均创造了我们1980年参加冬奥会以来的最佳成绩。label:体育
text:自然·大文学：在线发表了一项月球科学研究的重要成果。利用嫦娥五号月球“土特产”的同位素年龄和着陆区撞击坑统计结果，中科院天体信息创新研究院等单位的研究人员在目前常用月球年代函数的基础上建立了新的更精确的月球年代函数模型。label:教育
text:近日，教育部印发了《幼儿园保育教育质量评估指南》（以下简称《评估指南》）。教育部基础教育司负责人就《评估指南》有关内容回答了记者提问。label:房产
text:张涵予，1964年出生于北京，籍贯陕西蓝田，是中国内地男演员，1988年毕业于中央戏剧学院表演系，中国文艺界联合会第十届全委会委员，中国电影家协会副主席。在部队大院长大，读高中时，就到中央电视台译制片组给外国影片配音，1983年，张涵予成为专业配音演员。label:娱乐
text:甲醇期货今日挂牌上市继上半年焦炭、铅期货上市后，酝酿已久的甲醇期货将在今日正式挂牌交易。基准价均为3050元/吨继上半年焦炭、铅期货上市后，酝酿已久的甲醇期货将在今日正式挂牌交易。label:财经

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Figure 4: Validation / accuracy figure for Task B.

3 Results

3.1 Training and Validation Results

Tables 2 and 3 summarize the epoch-level results extracted from the training logs.

Table 2: Task A training and validation results.

Epoch	Train Loss	Train Acc	Val Loss	Val Acc
1	0.71	72.22%	1.00	70.08%
2	0.55	73.61%	0.88	73.48%
3	0.35	88.89%	0.85	76.16%
4	0.37	84.72%	0.79	78.04%
5	0.35	90.28%	0.83	77.36%
6	0.22	93.06%	0.82	78.52%

Table 3: Task B training and validation results.

Epoch	Train Loss	Train Acc	Val Loss	Val Acc
1	0.80	72.22%	1.00	70.28%
2	0.65	77.78%	0.85	74.32%
3	0.52	83.33%	0.86	75.72%
4	0.30	94.44%	0.77	78.16%
5	0.30	90.28%	0.73	79.32%
6	0.27	91.67%	0.80	78.04%

Task B reached a slightly higher best validation accuracy (79.32%) than Task A (78.52%). This difference already suggests that preprocessing may provide a small advantage.

3.2 Final Test Results

The most important assignment metrics are the final test loss and test accuracy. These are summarized in Table 4.

Table 4: Final comparison of Task A and Task B.

Task	Preprocessing	Test Loss	Test Accuracy
Task A	No	0.55	83.96%
Task B	Yes	0.52	84.20%

Task B improved the test accuracy by 0.24 percentage points and reduced the test loss from 0.55 to 0.52. The improvement is not large, but it is consistent with the validation trend and therefore meaningful in the context of this controlled comparison.

3.3 Misclassified Sample Comparison

To support the required qualitative analysis, the misclassified outputs of Task A and Task B were compared directly. In terms of the total number of errors on the test set, Task A produced 401 misclassified samples, while Task B produced 395 misclassified samples. This is also consistent with the reported test accuracies: under a 2,500-sample test set, 83.96% accuracy for Task A corresponds to about 401 errors, and 84.20% accuracy for Task B corresponds to about 395 errors.

A direct comparison of the two error lists further shows:

- 153 samples were misclassified in Task A but correctly classified in Task B;
- 147 samples were misclassified in Task B but correctly classified in Task A.

Therefore, Task B made fewer total mistakes overall, but the overlap analysis also shows that preprocessing does not help every headline. Some samples benefit from noise reduction, while other samples lose useful structural or formatting cues after punctuation removal.

3.4 Representative Cross-Case Examples

3.5 Representative Cross-Case Examples

To make the comparison clearer, six representative examples are selected below: three cases where Task A is wrong but Task B is correct, and three cases where Task A is correct but Task B is wrong. Preference is given to headlines whose surface form changes after preprocessing, because such cases better illustrate how punctuation removal or boundary flattening can influence prediction.

3.5.1 Task A wrong, Task B correct

Example 1. “调查显示：29.5%的人不满意当年所选高考专业”

Ground truth: Education. Task A predicted *Stocks*, while Task B was correct. This headline contains numbers and punctuation, which may distract Task A toward a more report-like or market-like pattern. After preprocessing, the model appears to focus more on the education-specific cues “高考” and “专业”, so Task B gives the correct label.

Example 2. “河南中招网上填报志愿：12日初三毕业生可确认”

Ground truth: Education. Task A predicted *Technology*, while Task B was correct. The phrase “网上填报” can easily attract a technology-related prediction in Task A. In Task B, removing punctuation makes the core admission context more continuous, so words such as “中招”, “志愿”, and “毕业生” become more salient as education signals.

Example 3. “热点观察：教育资源不足下的复读困境”

Ground truth: Education. Task A predicted *Stocks*, while Task B was correct. The prefix “热

点观察” gives the headline a generic commentary style, which may weaken the category signal in Task A. Task B benefits from the flatter text and captures the central education topic from “教育资源” and “复读” more directly.

3.5.2 Task A correct, Task B wrong

Example 1. “日本地震：金吉列关注在日学子系列报道”

Ground truth: Education. Task A was correct, while Task B predicted *Politics*. After preprocessing, the headline becomes “日本地震金吉列关注在日学子系列报道”. Removing the colon weakens the boundary between the background event and the real topic, so the disaster-related cue “日本地震” dominates more strongly and pushes Task B toward a politics-related interpretation.

Example 2. “公共英语 (PETS) 写作中常见的逻辑词汇汇总”

Ground truth: Education. Task A was correct, while Task B predicted *Game*. After preprocessing, the headline becomes “公共英语 PETS 写作中常见的逻辑词汇汇总”. The original form “(PETS)” is a clear exam-style marker, but removing the parentheses weakens this structured educational cue and makes the string look more like a loose keyword sequence, which hurts Task B.

Example 3. “在美国 “人肉搜索” 为何成重罪？(组图)”

Ground truth: Education. Task A was correct, while Task B predicted *Technology*. After preprocessing, the text becomes “在美国人肉搜索为何成重罪组图”. Removing quotation marks, the question mark, and parentheses makes “人肉搜索” stand out as a stronger internet-related expression, so Task B is pulled toward the technology category even though the original headline belongs to the education set.

4 Discussion

The comparison between Task A and Task B leads to two main conclusions. First, text preprocessing produced a small but consistent overall improvement. Task B achieved better validation performance, lower test loss, and higher final test accuracy. This suggests that removing punctuation and normalizing whitespace can reduce noise and make the core lexical information more accessible to the classifier.

Second, the misclassified examples show that preprocessing is not always beneficial. In some headlines, punctuation and surface structure preserve semantic boundaries or domain-specific expressions. Once these cues are removed, the resulting text may become less distinctive and may even increase the chance of confusion with another category. Therefore, preprocessing should be understood as a trade-off: it can remove noise, but it can also remove information.

Overall, the evidence suggests that lightweight preprocessing is worthwhile for this task, but its impact remains modest. More advanced preprocessing or model tuning would likely be

needed for a larger performance gain.

5 Conclusion

This report presented the required Task A and Task B experiments for Transformer-based Chinese news classification. Task A used the original text, while Task B applied lightweight preprocessing before training and testing. Based on the reported logs and generated outputs, Task B achieved a small improvement over Task A, raising test accuracy from 83.96% to 84.20% and reducing test loss from 0.55 to 0.52.

The comparison of misclassified samples further shows that preprocessing can help the model focus on more informative lexical cues, although it may also remove useful structural information in some cases. In summary, the experiment confirms that preprocessing is beneficial on average for this setting, but the effect is limited and should be interpreted carefully.